The suspension of a mere proposition is a set and its north-south path space is that proposition

The Univalent Foundations Program

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Editorial note. The counted statement is the author's prop:trunc_of_prop_is_set (source lines 537-540); the proof environment begins two lines below it (line 541 is blank), so the pairing is by adjacency.

Editorial note. The retained context is the author's own subsection heading with its label and the single sentence that announces the lemma, retained because the surrounding computation is used immediately afterwards for Diaconescu's theorem.

Editorial note. The retained counter values (chapter 10, section 1, subsection offset 4, theorem offset 12) reproduce the first-edition numbering: the heading prints as 10.1.5 and the counted lemma as Lemma 10.1.13.

Editorial note. The sethood criterion cited by the proof is located in hlevels.tex and is therefore declared as an external source pointer; its first-edition number is taken from the label index shipped beside the pinned source in this source directory.

Editorial note. No retained line refers to a figure, bibliography entry, included file or non-ASCII character, so the package declares no asset and no bibliography.

Editorial scope. One original result of the HoTT Book, counted once: the suspension of a mere proposition is a set and its north-south path space is that proposition.

Allowed standard prerequisites (editorial).

- Suspension as a higher inductive type. The suspension susp(A) of a type A is generated by north, south : susp(A) and merid : prd{a:A} north = south, with the matching induction and recursion principles.. The statement is about susp(A) and the proof defines a family P on susp(A) x susp(A) by recursion on the two suspensions and checks the meridian condition (source lines 543-571).
- The sethood criterion for reflexive mere relations. If P is a reflexive mere relation on a type X and there is tau : prd{x,y:X} P(x,y) -> (x = y), then X is a set and P(x,y) is equivalent to (x = y) for all x, y; this is book Theorem 7.2.2.. Applied twice, at source lines 573-575 to reduce the goal to constructing tau, and at line 610 to conclude sethood and

the equivalence of P with the identity type. The criterion lies in another chapter file and is declared as a non-load-bearing source pointer.

- Mere propositions and the unit type. An inhabited mere proposition is contractible and hence equivalent, and therefore equal by univalence, to the unit type.. Used in the meridian condition at source lines 563-569 to turn the four values of P into equal types, since the assumed inhabitant a : A makes A equivalent to unit.

Source reference. Original lemma 7.2.2: sources/hott-setmath/book-labelnumbers-first-edition.tex, source lines 1797–1797. The retained proof applies the sethood criterion for reflexive mere relations, which is proved in another chapter of the same book and is not part of this lemma. It is a navigation pointer to a standard prerequisite of the section: the counted lemma's own author proof is retained line for line, the criterion is not restated here, and no step of the counted proof is replaced by the pointer.

10.1.5 The axiom of choice implies excluded middle

We begin with the following lemma.

Lemma 10.1.13. *If A is a mere proposition then its suspension $\Sigma(A)$ is a set, and A is equivalent to $\mathsf{N} =_{\Sigma(A)} \mathsf{S}$.*

Proof. To show that $\Sigma(A)$ is a set, we define a family $P : \Sigma(A) \to \Sigma(A) \to \mathcal{U}$ with the property that $P(x, y)$ is a mere proposition for each $x, y : \Sigma(A)$, and which is equivalent to its identity type $\mathsf{Id}_{\Sigma(A)}$. We make the following definitions:

$$\begin{aligned} P(\mathsf{N},\mathsf{N}) &:\equiv \mathbf{1} & \qquad P(\mathsf{S},\mathsf{N}) &:\equiv A \\ P(\mathsf{N},\mathsf{S}) &:\equiv A & \qquad P(\mathsf{S},\mathsf{S}) &:\equiv \mathbf{1}. \end{aligned}$$

We have to check that the definition preserves paths. Given any $a : A$, there is a meridian $\mathsf{merid}(a) : \mathsf{N} = \mathsf{S}$, so we should also have

$$P(\mathsf{N},\mathsf{N}) = P(\mathsf{N},\mathsf{S}) = P(\mathsf{S},\mathsf{N}) = P(\mathsf{S},\mathsf{S}).$$

But since A is inhabited by a, it is equivalent to $\mathbf{1}$, so we have

$$P(\mathsf{N},\mathsf{N}) \simeq P(\mathsf{N},\mathsf{S}) \simeq P(\mathsf{S},\mathsf{N}) \simeq P(\mathsf{S},\mathsf{S}).$$

The univalence axiom turns these into the desired equalities. Also, $P(x, y)$ is a mere proposition for all $x, y : \Sigma(A)$, which is proved by induction on x and y, and using the fact that being a mere proposition is a mere proposition.

Note that P is a reflexive relation. Therefore we may apply original lemma 7.2.2, so it suffices to construct $\tau : \prod_{(x,y:\Sigma(A))} P(x, y) \to (x = y)$. We do this by a double induction. When x is N, we define $\tau(\mathsf{N})$ by

$$\tau(\mathsf{N},\mathsf{N},u) :\equiv \mathsf{refl}_{\mathsf{N}} \qquad \text{and} \qquad \tau(\mathsf{N},\mathsf{S},a) :\equiv \mathsf{merid}(a).$$

If A is inhabited by a then $\mathsf{merid}(a) : \mathsf{N} = \mathsf{S}$ so we also need $\mathsf{merid}(a)_*(\tau(\mathsf{N},\mathsf{N})) = \tau(\mathsf{N},\mathsf{S})$. This we get by function extensionality using the fact that, for all $x : A$,

$$\begin{aligned} \mathsf{merid}(a)_*(\tau(\mathsf{N},\mathsf{N},x)) = \tau(\mathsf{N},\mathsf{N},x) \cdot \mathsf{merid}(a)^{-1} \equiv \\ \mathsf{refl}_{\mathsf{N}} \cdot \mathsf{merid}(a) = \mathsf{merid}(a) = \mathsf{merid}(x) \equiv \tau(\mathsf{N},\mathsf{S},x). \end{aligned}$$

In a symmetric fashion we may define $\tau(\mathsf{S})$ by

$$\tau(\mathsf{S}, \mathsf{N}, a) :\equiv \mathsf{merid}(a)^{-1} \qquad \text{and} \qquad \tau(\mathsf{S}, \mathsf{S}, u) :\equiv \mathsf{refl}_{\mathsf{S}}.$$

To complete the construction of τ, we need to check $\mathsf{merid}(a)_*(\tau(\mathsf{N})) = \tau(\mathsf{S})$, given any $a : A$. The verification proceeds much along the same lines by induction on the second argument of τ.

Thus, by original lemma 7.2.2 we have that $\Sigma(A)$ is a set and that $P(x, y) \simeq (x = y)$ for all $x, y : \Sigma(A)$. Taking $x :\equiv \mathsf{N}$ and $y :\equiv \mathsf{S}$ yields $A \simeq (\mathsf{N} =_{\Sigma(A)} \mathsf{S})$ as desired. □